


Signal: antenna -> air-gap RC filter -> electrometer (G = 101) -> anti-alias + LMP7715 driver

-> PCM1804 (24 bit, 192 kHz) -> CS8406 -> S22083 -> AES3 110 R on RJ45 (STP cable, 100 m).

Supply: 6.98 V from the two-bucket PSU (or a 9-13 V battery) -> ADM7150 +5VA and +3V3.

Board: 200 x 100 mm, 4 layers, three compartments under an aluminium shield (layout.py).

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets

CERN-OHL-W-2.0

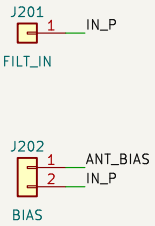
**ELARA -- ELF Atmospheric Radio Analyser**

Sheet: /
File: antenna_amplifier.kicad_sch

***Title: ELARA antenna amplifier***

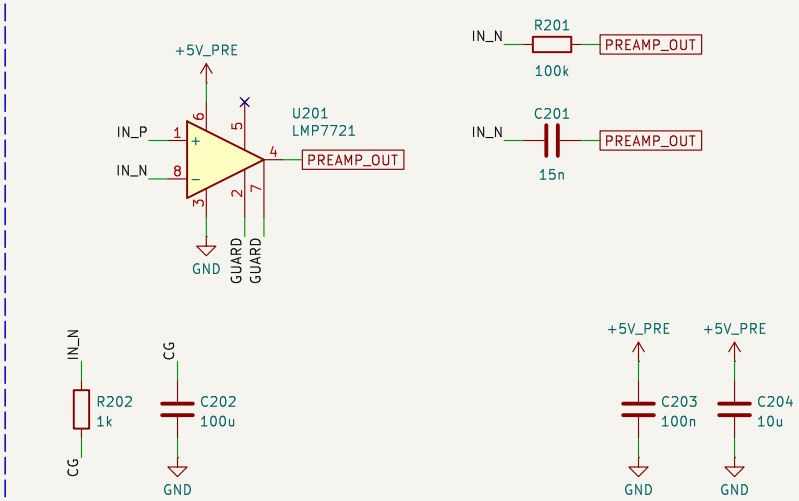
| Size: A4 | Date: 2026-09-30 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/5 |

## INPUT (comp. 1)

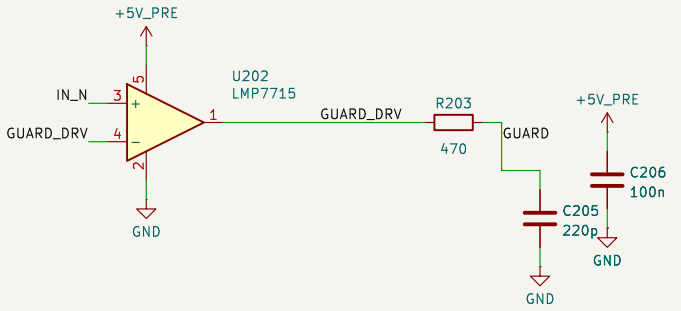



IN_P: guard on all layers,
no mask, PTFE turret.
J202: G-ohm R, no header.

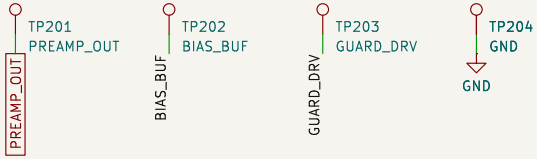
## ELECTROMETER G = 101



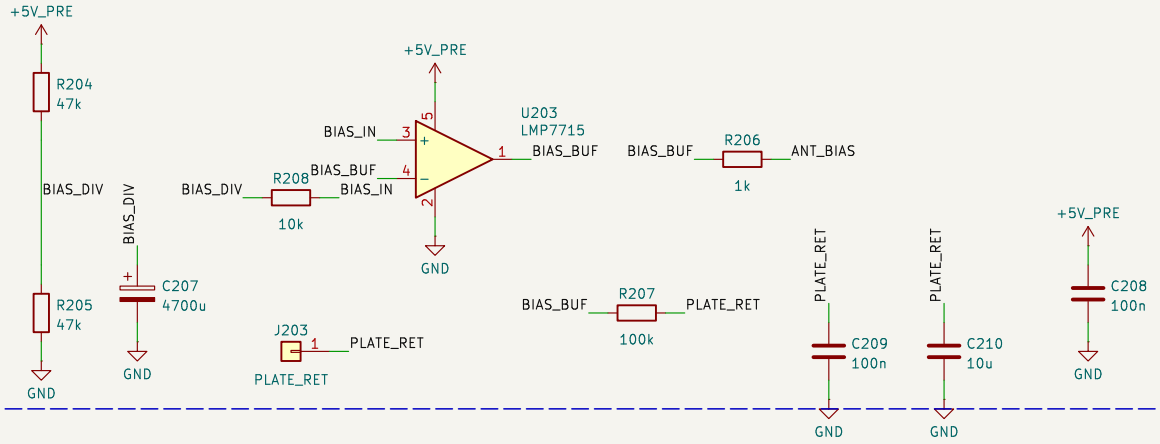

## GUARD DRIVER




## TEST POINTS




## ANTENNA BIAS 2.5 V



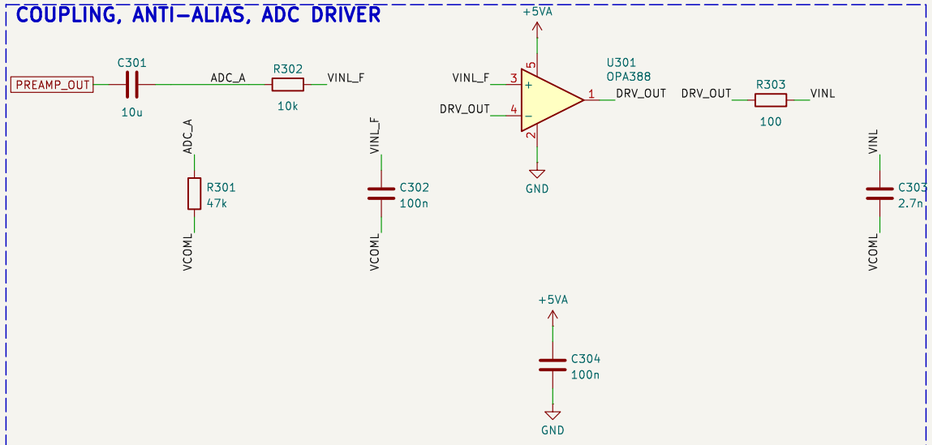

Cg returns to GND: the 2.5 V buffer noise stays out of the gain path.
Guard buffer senses IN− (virtual short to IN+): no bias current on IN_P.

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CERN-OHL-W-2.0
**ELARA -- ELF Atmospheric Radio Analyser**
Sheet: /Electrometer front end/
File: frontend.kicad_sch

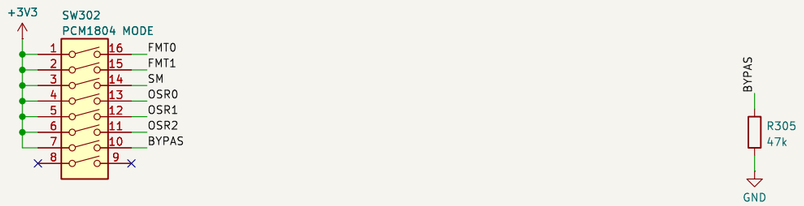
***Title: ELARA antenna amplifier — Electrometer front end***

| Size: A4 | Date: 2026-09-30 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/5 |

## COUPLING, ANTI-ALIAS, ADC DRIVER

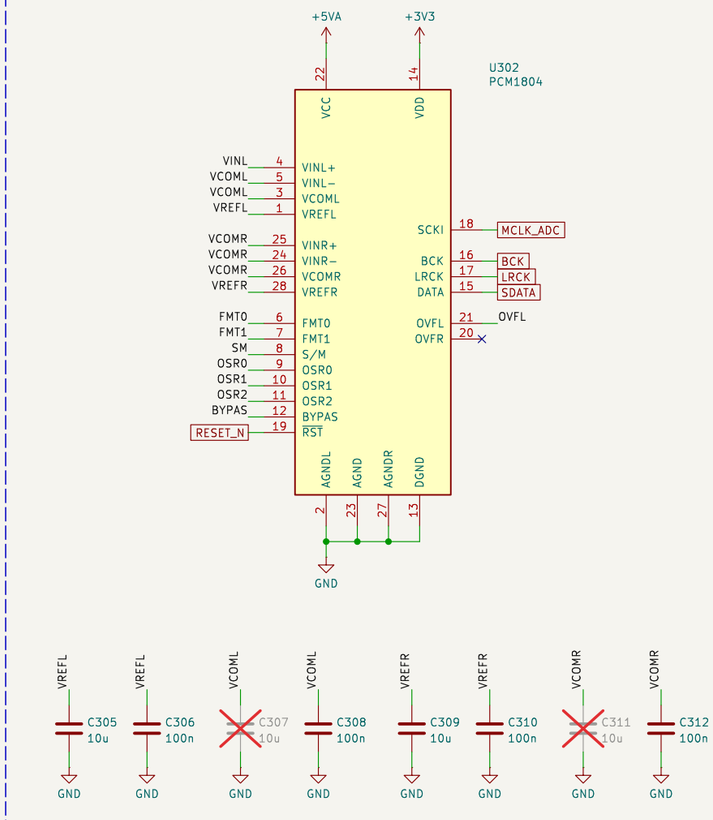



## PCM1804 MODE (ON = 1)



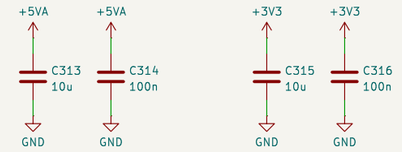

FMT0..OSR2: internal 51 k pull-downs.

## PCM1804 master, 192 kHz




Default (192 kHz): S/M=0 master (switch 3 OFF), OSR2..0=111 (SCKI=128 fs), FMT1..0=01 I2S, BYPAS=1 (HPF off).
Change switches with power off, or press RESET afterwards.

## SUPPLY DECOUPLING

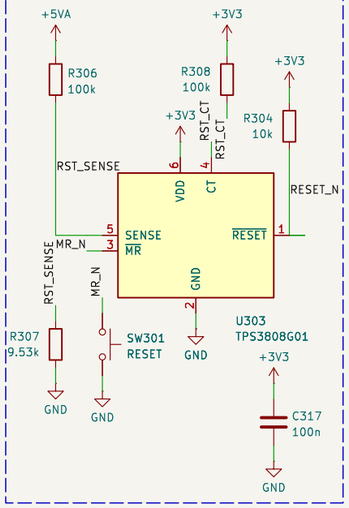



## OVERLOAD FLAG




No LED (battery use):
meter or scope on TP301.

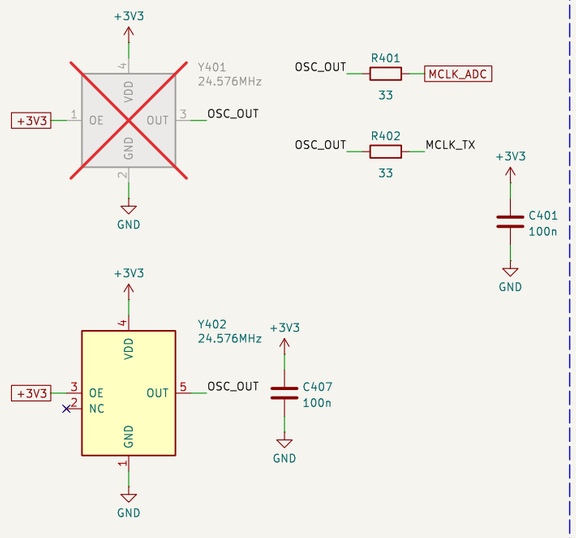
## RESET supervisor





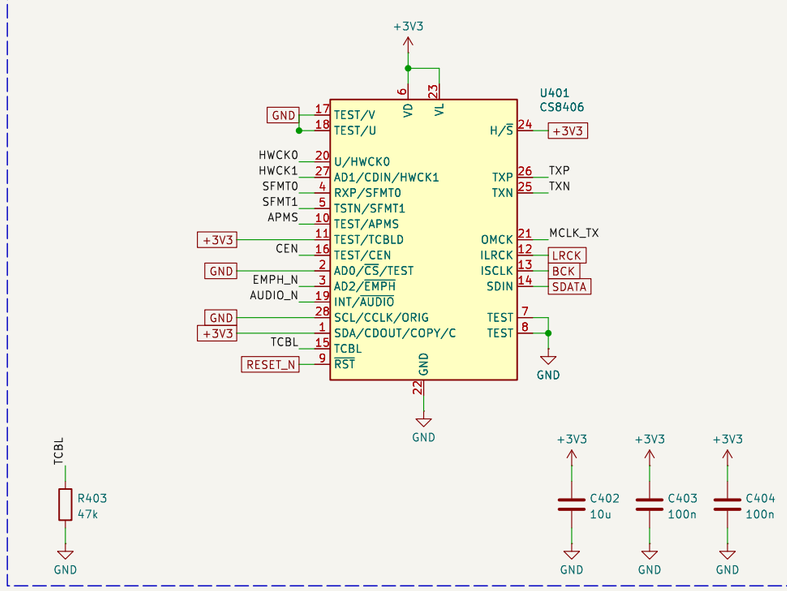
Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0
**ELARA -- ELF Atmospheric Radio Analyser**
Sheet: /ADC (PCM1804)/
File: adc.kicad_sch
***Title: ELARA antenna amplifier — ADC (PCM1804)***
Size: A3 | Date: 2026-09-30 | **Rev: 0.2**
KiCad E.D.A. 9.0.9 | Id: 3/5

## MASTER CLOCK 24.576 MHz

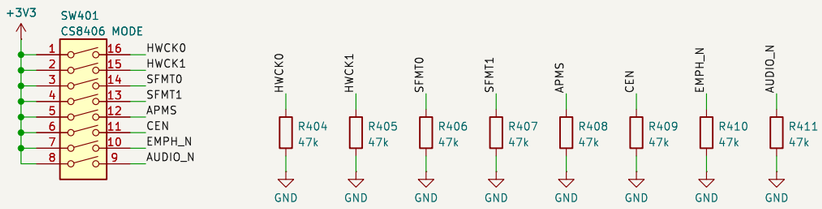



Fit Y402 (SOT23-5, leaded) or Y401 (3.2x2.5 no-lead), never both.

## CS8406 hardware mode




## CS8406 MODE (ON = 1)



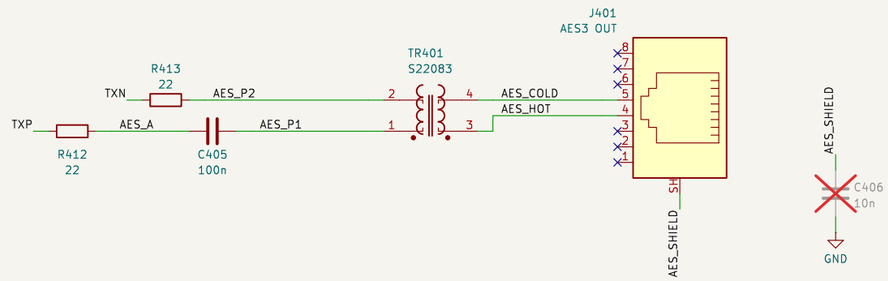

Default (192 kHz): HWCK1..0=01 (switch 2 OFF) -> OMCK = 128 fs, SFMT = I2S,
APMS=0 (slave), CEN=0, EMPH_N=1, AUDIO_N=0 (data sheet Table 4: 11 is 256 fs).

## TEST POINTS (clocks, reset, digital GND)




## AES3 OUTPUT transformer isolated, RJ45 for shielded twisted pair

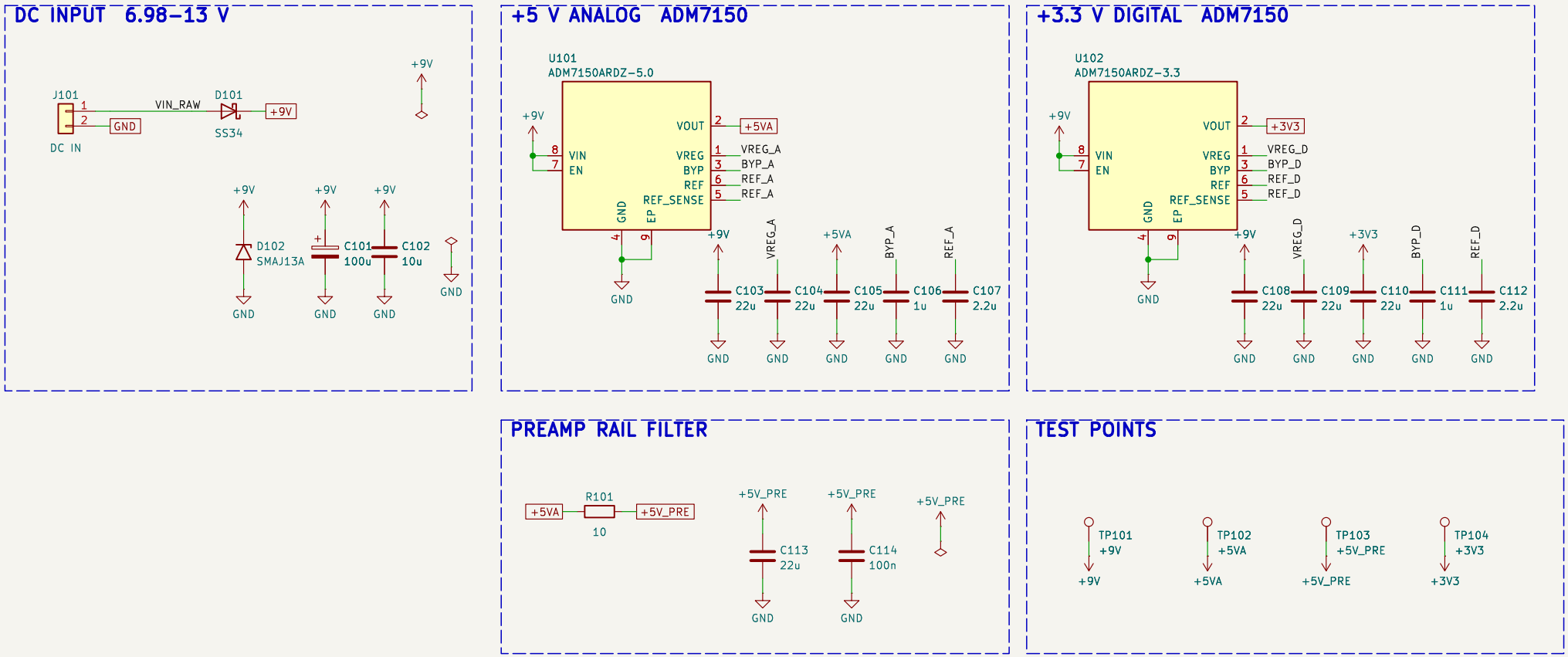
**AES3 110 R balanced**




AES3: 2 x 22 R + 2 x 33.5 R driver (CS8406 at VL = 3.3 V) = 111 R source, 3.3 Vpp into 110 R.
Cable: shielded Cat5e/6 (100 R pairs), shield connected at the indoor end only. AES3 is polarity-free.

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CERN-OHL-W-2.0
**ELARA -- ELF Atmospheric Radio Analyser**

| | | |
|---|---|---|
| Sheet: /Clock and AES3 output/ | | |
| File: digital.kicad_sch | | |
| ***Title: ELARA antenna amplifier — Clock and AES3 output*** | | |
| Size: A3 | Date: 2026-09-30 | **Rev: 0.2** |
| KiCad E.D.A. 9.0.9 | | Id: 4/5 |




Budget: ~42 mA on +5VA, ~53 mA on +3V3 (15 mA of it AES3 line current), ~100 mA at J101.

ADM7150: REF_SENSE tied to REF (fixed output), EN tied to VIN.

No switching regulators on this board. +9V is the DC bus after D101 (about 6.0 V from the PSU, up to 13 V from a battery); the net keeps its rev 0.1 name.

No LEDs: the board may run from a battery. Check the rails at the test points with a voltmeter.

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CERN-OHL-W-2.0

**ELARA -- ELF Atmospheric Radio Analyser**

Sheet: /Power input and regulators/

File: power.kicad_sch

***Title: ELARA antenna amplifier — Power input and regulators***

| Size: A4 | Date: 2026-09-30 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 5/5 |